



UNIVERSITI MALAYSIA PERLIS

MINI PROJECT REPORT

NDJ20403 MICROCONTROLLER SYSTEM

Title : Digital Lock Security System

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1.0 Introduction

1.1 Background of the Project

In recent years, security systems have become increasingly important in both residential and commercial environments. Traditional locking mechanisms that rely on physical keys are gradually being replaced by digital systems due to their higher reliability and reduced risk of unauthorized access. Digital lock systems provide improved control, flexibility and user interaction.

This project presents the design and development of a digital lock security system using an STC 8051 microcontroller. The system integrates multiple hardware components, including an infrared (IR) sensor, keypad, LCD display, LEDs, and a buzzer, to form a complete and functional access control system. The IR sensor is used to detect user presence and activate the system, while the keypad allows users to input a password for authentication.

The microcontroller processes the entered password and determines whether access should be granted or denied. The system also provides appropriate feedback through both visual and audible indicators. The LCD displays system messages and instructions, while the LEDs indicate different system states such as access granted, access denied, and system lock. In addition, the buzzer is activated during security events to provide an audible alert.

Overall, this project demonstrates how an embedded system can be effectively utilized to enhance security through automation, real-time processing, and intelligent control.

1.2 Objective and Scope of the Project

The main objective of this project is to design and implement a microcontroller-based digital lock system using the 8051 platform. The system aims to provide a secure password authentication mechanism that allows only authorized users to gain access. In addition, the system is designed to limit the number of incorrect password attempts in order to enhance security and prevent unauthorized access through repeated trials.

Furthermore, the project seeks to provide clear and effective system feedback through both visual and audible indicators. The LCD display is used to present system messages and instructions, while LEDs indicate different system states such as access granted, access denied, and system lock. The buzzer is included to generate an audio alert during security events. Another important objective of the system is to implement an administrator control function that allows the system to be reset after it has been locked due to multiple incorrect attempts. Additionally, the system is designed to track the number of successful door unlock operations, providing basic monitoring functionality.

In order to enhance system performance and efficiency, the project also incorporates the use of microcontroller peripherals and interrupt mechanisms. A timer peripheral is utilized to perform time-based operations such as delay generation and inactivity detection. This allows the system to automatically return to idle mode after a period of no user interaction. In addition, a timer interrupt is implemented to handle time tracking in a non-blocking manner, ensuring that the system can continue processing user input while maintaining accurate timing control.

The scope of this project focuses on the development of a basic embedded digital lock system using discrete hardware components and the 8051 microcontroller. The system includes user detection through an infrared sensor, password input via a keypad, and access control through password verification. It also provides system feedback through LEDs, an LCD display, and a buzzer. The implementation is limited to a standalone system without external communication or networking capabilities, and it is intended primarily for educational purposes and small-scale security applications.

1.3 Limitations

Despite the successful implementation of the system, several limitations have been identified. The user and admin passwords are fixed and cannot be changed dynamically, which means they cannot be changed dynamically without reprogramming the microcontroller. This reduces the flexibility of the system and limits its applicability in real-world scenarios where password updates are required.

Another limitation is the system does not include non-volatile memory such as EEPROM, meaning the door open count will reset when power is turned off. This restricts the system's ability to perform long-term monitoring and data logging.

In addition, the system relies on keypad-based input, which provides moderate security compared to more advanced technologies such as biometric authentication. Environmental factors may also affect the performance of the IR sensor.

Another important limitation is related to the use of the Keil evaluation version, which imposes a code size restriction of 2KB. Due to this constraint, the program must be carefully optimized to minimize memory usage, and certain advanced features such as enhanced user interfaces, additional security mechanisms, or extended functionalities could not be implemented. This limitation directly influenced the overall system design and functionality.

Furthermore, the system makes use of software-based delay for certain operations, which may reduce efficiency compared to fully interrupt-driven designs. Although a timer interrupt is implemented, some parts of the system still rely on delay loops, which may affect responsiveness under more complex conditions.

2.0 Methodology

2.1 Equipment

The following hardware components were used in the development and implementation of the digital lock security system:

- STC89C52 microcontroller (8051-based)
- Infrared (IR) Sensor
- 5 x 4 matrix keypad
- LCD display module (I2C)
- Purple LED
- Traffic light LED module (Red, Yellow, Green LEDs)
- Buzzer
- Resistor 1k Ω

2.2 Block Diagram

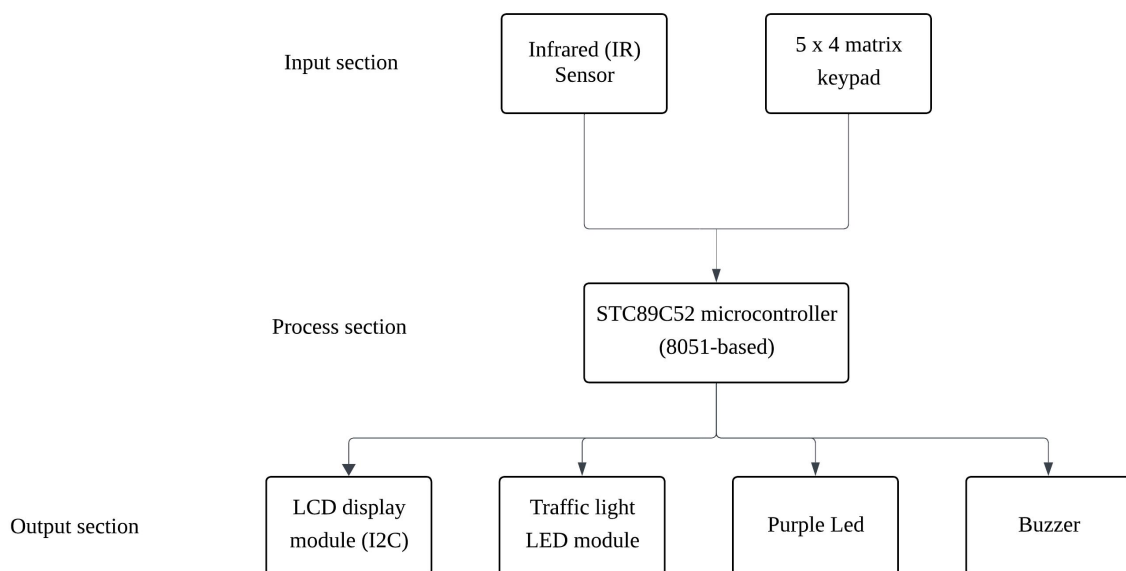


Figure 1.1: Block Diagram of the Digital Lock Security System

The digital lock system is composed of three main sections: the input section, processing section, and output section. Each section performs a specific role in ensuring the overall functionality of the system. A block diagram illustrating the overall system architecture is shown in Figure 1.1.

Input Section:

The input section consists of an infrared (IR) sensor and a keypad. The IR sensor is used to detect the presence of a user and automatically activate the system from idle mode, improving system efficiency and user interaction. The keypad serves as the primary input device, allowing users and administrators to input passwords and commands.

Processing Section:

The processing section is handled by the STC89C52 microcontroller, which is based on the 8051 architecture and acts as the main controller, processing all input signals, comparing passwords, managing attempts, and controlling system operations. The microcontroller also coordinates the interaction between input and output components to ensure proper system operation.

Output Section:

The output section includes an LCD display, LEDs, and a buzzer. The LCD display provides system messages such as password prompts and status updates. The LEDs indicate system conditions where the green LED represents access granted, the yellow LED represents incorrect password entry, the red LED represents system lock, and the purple LED indicates that the system is active. The buzzer generates an audible warning signal during security events, enhancing the overall security of the system.

2.3 System Operation

The system initially operates in standby mode, where the LCD displays “SYSTEM READY” on the first line and “WAITING...” on the second line to indicate that the system is idle and waiting for user interaction. When the infrared (IR) sensor detects the presence of a user, the system transitions to active mode. The LCD first displays “WELCOME” followed by “PLEASE ENTER”, and then updates to “ENTER PASSWORD” to prompt the user for input.

The user enters the password using the keypad, and each input character is masked with an asterisk (*) on the display to ensure security. The predefined password for normal access is “1234”. If the entered password matches the stored password, the system grants access by displaying “ACCESS GRANTED” on the LCD, activating the green LED, and incrementing the door open counter.

If the entered password is incorrect, the system displays “ACCESS DENIED” on the first line and “TRY LEFT:” followed by the remaining number of attempts on the second line. The yellow LED is activated to indicate an error condition. The system allows a maximum of three attempts.

After three consecutive incorrect attempts, the system enters a locked state. In this condition, the LCD displays “SYSTEM LOCKED” on the first line and “CONTACT ADMIN” on the second line. The red LED is activated, and the buzzer generates a timed audible warning signal. Normal user access is disabled at this stage.

To reset the system, the administrator must first enter admin mode using a designated key. Once in admin mode, the LCD displays “ADMIN PASSWORD” to prompt for the administrator input. The administrator then enters the password “8888*”. If the correct admin password is entered, the system resets successfully, and the LCD displays “SYSTEM RESET” before returning to the normal password entry screen with “ENTER PASSWORD”. The red LED is turned off, and normal operation resumes.

If an incorrect admin password is entered, the system triggers a warning condition. During this state, the LCD alternates between displaying “ADMIN FAILED” and then update to first displays “SYSTEM LOCKED followed by WAIT 30 SEC”. At the same time, the red LED blinks and the buzzer generates repeated audible signals for approximately 30 seconds. After this period, the administrator is allowed to attempt the reset again.

In addition to the main operation, the system includes a status display feature. When the user presses the designated key, the LCD displays “SYSTEM STATUS” on the first line and “DOOR OPEN:” followed by the number of successful access operations on the second line.

Furthermore, if no activity is detected for a certain period, the system triggers a timeout condition. The LCD displays “TIME OUT” followed by “TRY AGAIN” before returning to the standby LCD displays “SYSTEM READY” on the first line and “WAITING...” on the second line. This ensures efficient system operation and enhances overall security.

The overall system operation can be illustrated using a flowchart, as shown in Figure 2.2.

2.3.1 LCD Display Messages and System State Table

To provide a clearer and more structured understanding of the system behavior, the relationship between system states, user actions, LCD display messages, and system responses is summarized in Table 1.

System State	User Action / Condition	LCD Display (Line 1 / Line 2)	System Output / Response
Standby Mode	No user detected	SYSTEM READY / WAITING...	System idle
Activation	IR sensor detects user	WELCOME / PLEASE ENTER → ENTER PASSWORD	System activated
Password Entry	User inputs password	****	Input masked
Access Granted	Correct password	ACCESS GRANTED	Green LED ON, counter +1
Access Denied	Incorrect password	ACCESS DENIED / TRY LEFT: X	Yellow LED ON
Locked State	3 incorrect attempts	SYSTEM LOCKED / CONTACT ADMIN	Red LED ON, buzzer ON
Admin Mode	Admin key pressed	ADMIN PASSWORD	Waiting for admin input
Admin Success	Correct admin password	SYSTEM RESET → ENTER PASSWORD	System reset
Admin Failure	Incorrect admin password	ADMIN FAILED / SYSTEM LOCKED	Alarm triggered
Alarm State	Intruder condition	SYSTEM LOCKED / WAIT 30 SEC	LED blinking + buzzer
Status Display	Status key pressed	SYSTEM STATUS / DOOR OPEN: XX	Display counter
Timeout	No activity	TIME OUT / TRY AGAIN	Return to standby

Table 1: System State, Action and Output Representation

2.4 Flowchart

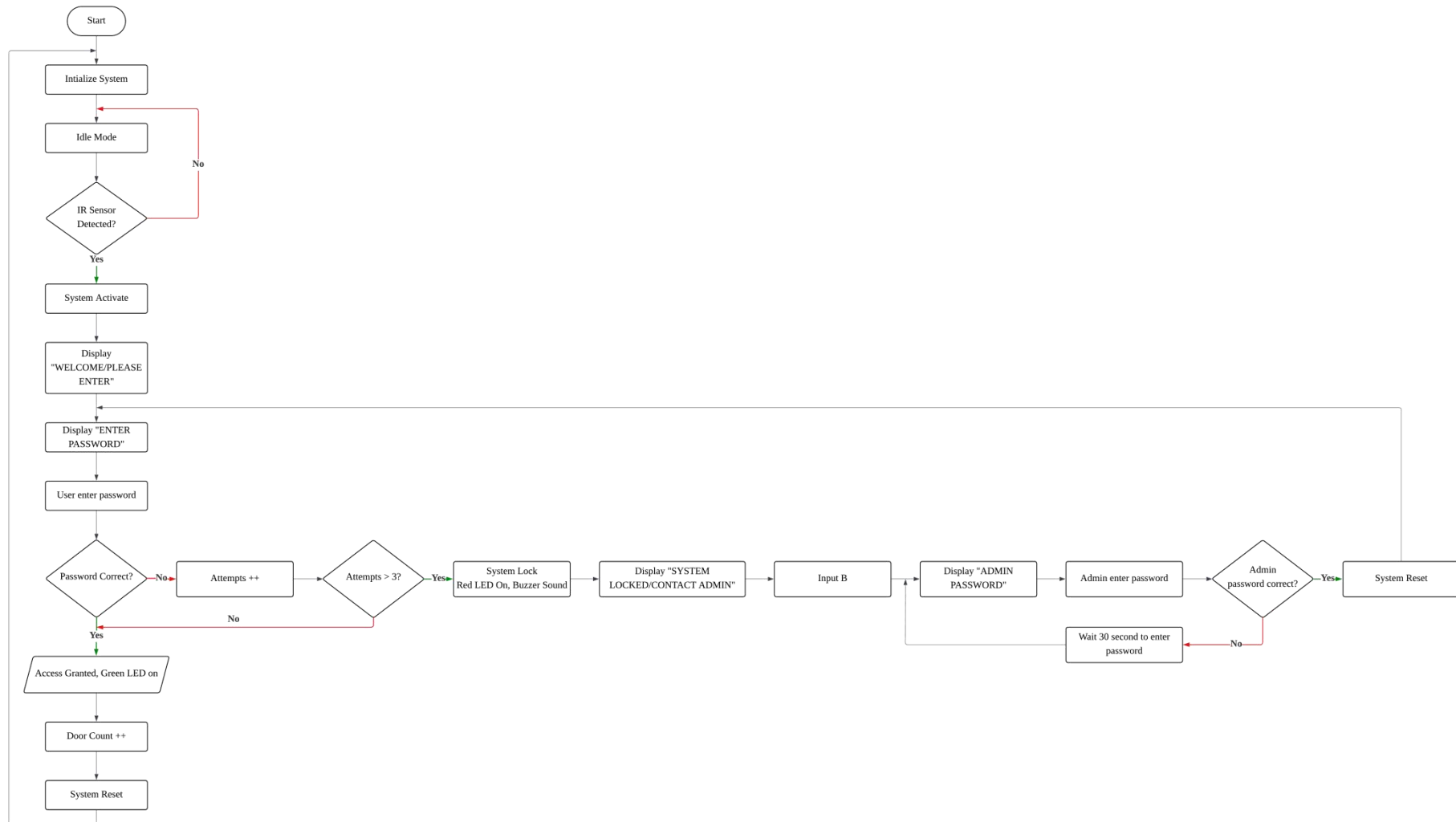


Figure 1.2: Flowchart of the Digital Lock Security System Operation

2.5 Explanation of Key Code Segment

This section explains the key parts of the program that control the operation of the digital lock security system. The implementation is organized into several modules, including input handling, output control, peripheral usage, interrupt handling, and system logic. These components work together to ensure reliable and efficient system performance.

2.5.1 Input Port

The system uses two main input devices, which are a keypad and an infrared (IR) sensor. These inputs allow user interaction and automatic system activation.

The keypad is connected to Port 2 and partially to Port 3 of the microcontroller. It is used for entering passwords and commands. The scanning process is implemented in the *keypad_scan()* function, where each row is activated and the columns are checked to detect which key is pressed.

```
char keypad_scan()
{
    char row;

    R1=R2=R3=R4=R5=1;

    for(row=0;row<5;row++)
    {
        R1=R2=R3=R4=R5=1;

        if(row==0)R1=0;
        if(row==1)R2=0;
        if(row==2)R3=0;
        if(row==3)R4=0;
        if(row==4)R5=0;

        if(C1==0){delay_ms(20);while(C1==0);return keypad[row][0];}
        if(C2==0){delay_ms(20);while(C2==0);return keypad[row][1];}
        if(C3==0){delay_ms(20);while(C3==0);return keypad[row][2];}
        if(C4==0){delay_ms(20);while(C4==0);return keypad[row][3];}
    }

    return 0;
}
```

A short delay is included to eliminate mechanical bouncing, and the function ensures that the key is released before accepting another input. This improves the accuracy and stability of keypad input detection. The matrix configuration also reduces the number of required I/O pins, making it an efficient design choice.

In addition, an IR sensor is connected to pin P3.2. This sensor detects the presence of a person and activates the system when triggered. The activation condition is implemented in the main program as follows:

```
if(IR_SENSOR==0 && system_active==0)
{
    system_active=1;
    idle_time=timer_tick;
    idle_displayed=0;

    timeout_flag = 0;
    admin_required = 0;

    admin_mode = 0;

    LED_PURPLE = 1;

    lcd_send_cmd(0x01);
```

This ensures that the system only starts when motion is detected, improving efficiency.

2.5.2 Output Port

The system includes several output devices such as LEDs, an LCD display, and a buzzer, all controlled by the microcontroller.

The LEDs are connected to Port 1 and are used to indicate different system states. For example, the green LED indicates successful access, while the red LED indicates that the system is locked.

```
LED_GREEN = 1; // Access granted  
LED_RED = 1; // System locked  
LED_YELLOW = 1; // Warning
```

The LCD display is connected using an I2C interface and is used to display system messages. The function `lcd_write_line()` is used to show text on the LCD.

```
lcd_write_line("ENTER PASSWORD",1);  
lcd_write_line("ACCESS GRANTED",1);  
lcd_write_line("SYSTEM LOCKED",1);
```

These messages provide real-time feedback to the user, improving usability and interaction with the system.

The buzzer is used as an alarm output and is activated during security events such as unauthorized access or system lock. The buzzer control is implemented using the following function:

```
void buzzer_alarm()  
{  
    unsigned int i;  
    BUZZER=0;  
    for(i=0;i<50;i++) delay_ms(100);  
    BUZZER=1;  
}
```

This produces a continuous sound to alert users during unauthorized access.

2.5.3 Peripheral

The project uses Timer0 of the 8051 microcontroller as a peripheral for timing and delay management. The timer is initialized in the *timer0_init()* function.

```
void timer0_init()
{
    TMOD |= 0x01;
    TH0=0x3C;
    TL0=0xB0;
    ET0=1;
    EA=1;
}
```

The timer continuously runs in the background and updates the variable *timer_tick*, which is used for time tracking. This allows the system to implement features such as automatic timeout.

```
if(timer_tick - idle_time > 200 && !locked)
{
    system_active = 0;
}
```

This ensures the system resets after a period of inactivity.

2.5.4 Interrupt Source

An interrupt mechanism is implemented using the Timer0 interrupt. This allows the system to handle time-based operations efficiently without stopping the main program.

The interrupt service routine (ISR) is defined as follows:

```
void timer0_ISR(void) interrupt 1
{
    TH0=0x3C;
    TL0=0xB0;
    timer_tick++;
}
```

Each time the timer overflows, the interrupt service routine (ISR) is triggered automatically, and the *timer_tick* variable is incremented. This provides a reliable time reference for the system. The use of interrupt-based timing is more efficient compared to delay-based implementation, as it allows the microcontroller to continue executing other tasks while maintaining accurate time tracking.

2.5.5 System State Control

The system uses multiple flag variables such as *system_active*, *locked*, *admin_mode*, and *admin_required* to control different operating states.

```
if(attempts>=max_attempt)
{
    locked=1;
    admin_required = 1;

    LED_RED=1;

    lcd_send_cmd(0x01);
    lcd_write_line("SYSTEM LOCKED",1);
    lcd_write_line("CONTACT ADMIN",2);

    buzzer_alarm();

    clear_input();
    index = 0;
    cursor_pos = 0;
}
```

These flags determine how the system behaves under different conditions. For example, when the number of incorrect attempts reaches the maximum limit, the system enters a locked state and disables normal user access.

This implementation allows the system to behave like a finite state machine (FSM), where it transitions between standby, active, locked, and admin modes.

2.5.6 Password Storage and Input Handling

The system stores the predefined password in an array, while user input is stored in a separate buffer.

```
char password[4]={'1','2','3','4'};  
char input[6];
```

Each key press is stored sequentially using an index counter.

```
input[index++]=key;
```

The entered password is compared with the stored password character by character.

```
for(i=0;i<4;i++)  
if(input[i]!=password[i]) break;
```

This ensures accurate authentication. If all characters match, access is granted; otherwise, the system records a failed attempt.

2.5.7 LCD Communication Control

The LCD module is controlled using I2C communication, which reduces the number of required I/O pins.

```
void lcd_write_line(char *str,unsigned char line)
{
    if(line==1)
        lcd_send_cmd(0x80);
    else
        lcd_send_cmd(0xC0);

    lcd_send_str(str);
}
```

The system sends commands and data in two 4-bit segments to control the LCD. This modular design simplifies display handling and improves code readability.

Overall, the code demonstrates a well-structured embedded system design that integrates input handling, output control, peripheral usage, and interrupt mechanisms. The use of flag-based state control allows the system to operate as a finite state machine, improving clarity and reliability. However, some functions such as the buzzer and alarm routines rely on delay-based implementation, which may temporarily block system execution and reduce responsiveness. Future improvements could include the use of fully interrupt-driven or non-blocking designs to enhance system performance.

3.0 Implementation & Results Discussion

3.1 Test Procedure and Result

The digital lock security system was implemented by integrating all hardware components on a breadboard and programming the STC89C52 microcontroller using C language. The complete hardware setup is shown in Figure 2.1, where all modules including the keypad, IR sensor, LCD display, LEDs, buzzer, and microcontroller are properly interconnected. The microcontroller section is further illustrated in Figure 2.2(a), which shows the overall setup on the breadboard, while Figure 2.2(b) provides a closer view of the STC89C52 microcontroller chip that serves as the main controller responsible for processing inputs and controlling outputs.

The keypad interface, as presented in Figure 2.3, demonstrates the row and column connections used for scanning user input. This matrix configuration allows efficient detection of key presses while minimizing the number of required I/O pins, indicating an effective and optimized hardware design. Meanwhile, the LCD module connected via the I2C interface is shown in Figure 2.4, where the SDA and SCL lines enable communication between the microcontroller and the display. This approach reduces wiring complexity and improves overall system integration.

The IR sensor module, shown in Figure 2.5, is used to detect the presence of a user. When motion is detected, the system is automatically activated. This confirms that the sensor-based triggering mechanism operates correctly and enhances system efficiency by preventing unnecessary operation when no user is present. The LED indicators in Figure 2.6 provide visual feedback for different system states, while the buzzer module in Figure 2.7 functions as an audible alert during security events.

During operation, the system initially remains in standby mode, as shown in Figure 2.8, where the LCD displays "SYSTEM READY." When a user is detected, the system transitions to active mode and prompts for password entry. The password input process is displayed as masked characters, as illustrated in Figure 2.9, ensuring that the entered password is not visible and improving system security.

When the correct password is entered, the system grants access successfully, as shown in Figure 2.10, together with the activation of the green LED. This confirms that the password verification logic and authentication mechanism are functioning correctly. Conversely, when an incorrect password is entered, the system displays an “ACCESS DENIED” message, as seen in Figure 2.11, along with the activation of the yellow LED. This demonstrates that the system can accurately detect invalid input and provide appropriate feedback to the user.

After three consecutive incorrect attempts, the system enters a locked state, as illustrated in Figure 2.12, where further access is restricted. In this condition, the system triggers an alarm sequence. As shown in Figure 2.13(a), the LCD displays “ADMIN FAILED” when an incorrect administrator password is entered. Subsequently, Figure 2.13(b) shows the message “SYSTEM LOCKED / WAIT 30 SEC,” indicating that the system enforces a delay before allowing another attempt. At the same time, the buzzer and red LED are activated, providing both audible and visual alerts. This confirms that the security mechanism is effective in preventing repeated unauthorized access attempts.

Finally, the system recovery process is demonstrated in Figure 2.14, where the administrator enters the correct reset password. This action successfully restores the system to its normal operating state, indicating that the admin control function is working as intended.

Overall, the test results demonstrate that the system operates reliably under different operating conditions. All components function together effectively to achieve the intended objectives of the project. The integration of sensor-based activation, password authentication, and alarm mechanisms enhances both system usability and security. However, it is observed that certain operations, particularly the buzzer and alarm routines, rely on delay-based implementation, which may temporarily block system execution and reduce responsiveness. Future improvements could focus on implementing non-blocking or interrupt-driven approaches to further enhance system performance.

3.2 Hardware Photos and Explanation

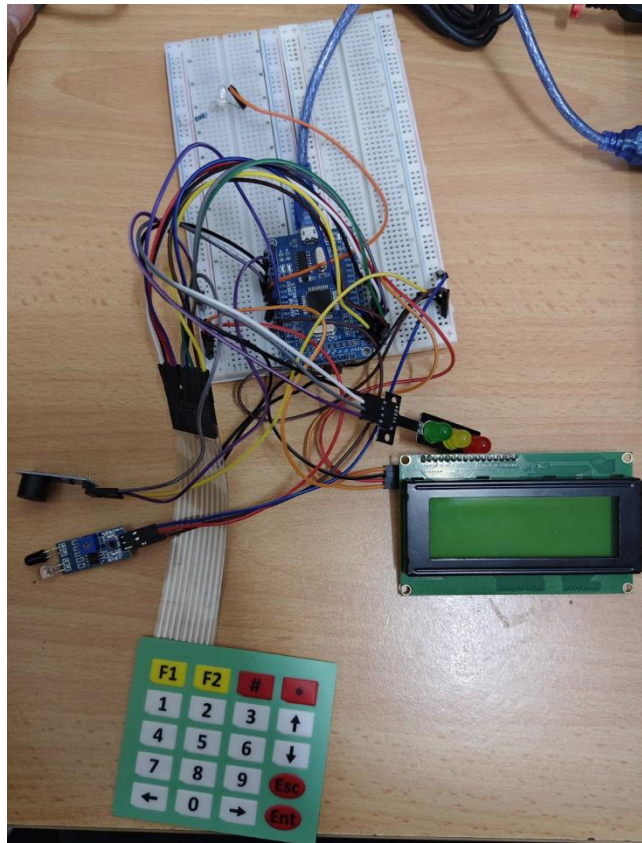


Figure 2.1: Overall Hardware Setup

This figure shows the complete system with all components interconnected on a breadboard, demonstrating proper integration and wiring of the entire system.

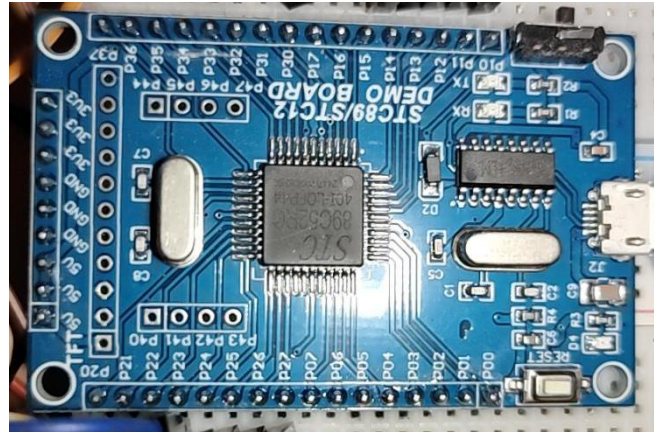


Figure 2.2(a): STC89C52 Microcontroller

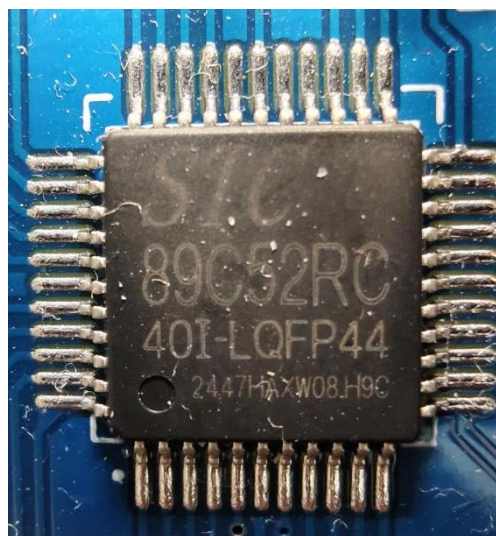


Figure 2.2(b): STC89C52 Microcontroller Chip

Figure 2.2(a) shows the overall microcontroller setup on the breadboard, including the connections required for system operation. Figure 2.2(b) provides a closer view of the STC89C52 microcontroller chip, which acts as the main controller responsible for processing inputs and controlling outputs.



Figure 2.3: Keypad Interface

This figure shows the row and column connections used for matrix keypad scanning, enabling efficient input detection.

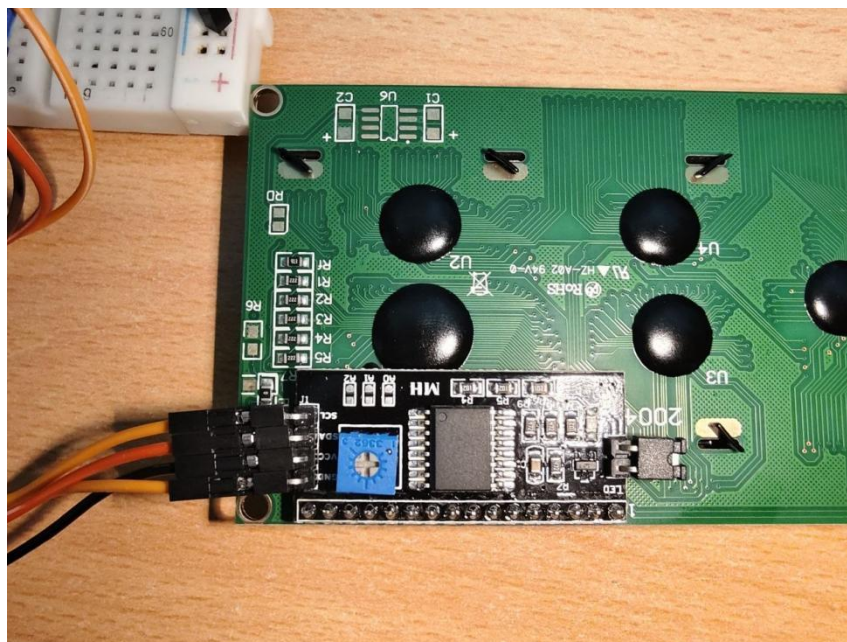


Figure 2.4: LCD with I2C Module

This figure shows the LCD module connected via I2C communication using SDA and SCL lines, reducing wiring complexity.

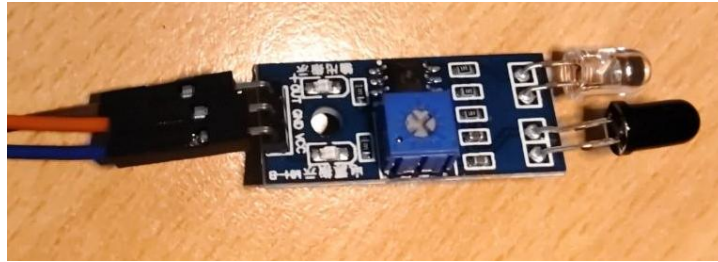


Figure 2.5: IR Sensor Module

This figure shows the infrared sensor used to detect user presence and activate the system automatically.

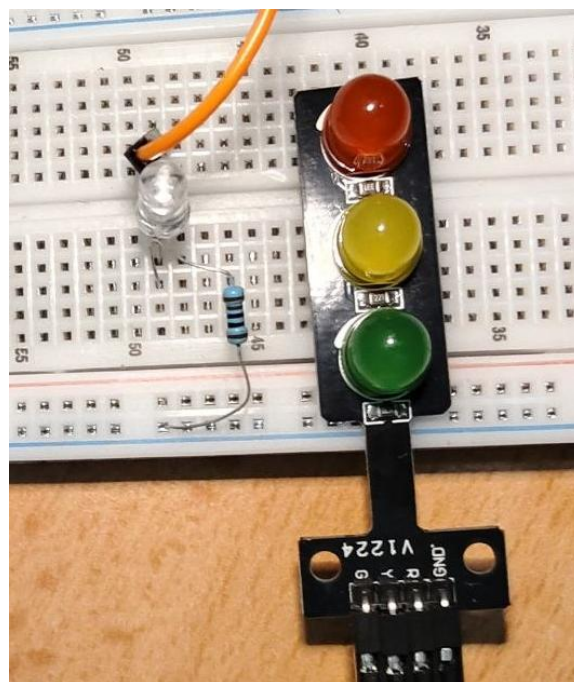


Figure 2.6: LED Indicators

This figure illustrates the LED indicators used to represent different system states such as access granted, denied, and locked.



Figure 2.7: Buzzer Module

This figure shows the buzzer used as an alarm output device during security events.



Figure 2.8: Idle Mode Display

This figure shows the system in standby mode with the LCD displaying “SYSTEM READY.”.



Figure 2.9: Password Entry Display

This figure shows the password input process where characters are masked for security.



Figure 2.10: Access Granted

This figure shows successful authentication with the corresponding system response.



Figure 2.11: Access Denied

This figure shows the system response when an incorrect password is entered.



Figure 2.12: System Locked

This figure shows the system entering a locked state after multiple failed attempts.



Figure 2.13(a): Admin Failed Display Message

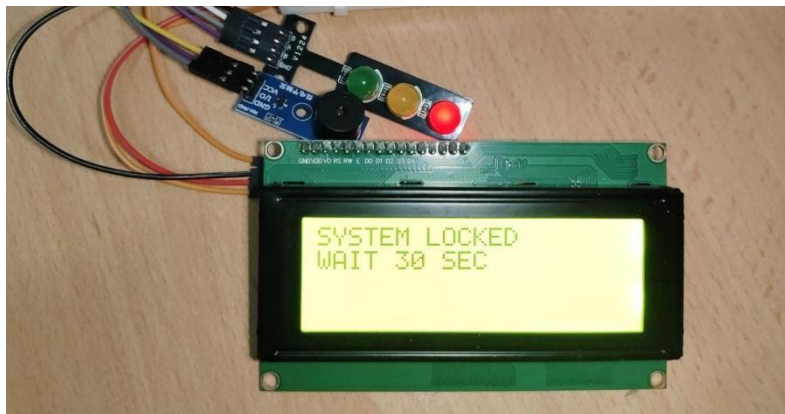


Figure 2.13: Intruder Alarm

Figure 2.13(a) shows the display message “ADMIN FAILED,” which appears when an incorrect administrator password is entered. Figure 2.13(b) shows the system warning message “SYSTEM LOCKED / WAIT 30 SEC,” indicating that the system is in a locked state and the user must wait before attempting again.



Figure 2.14: Admin Reset

This figure shows the system recovery process after successful administrator authentication.

4.0 Conclusion

In conclusion, this project successfully achieved the design and implementation of a smart door security system using the STC89C52 (8051-based) microcontroller. The system effectively integrates multiple hardware components, including a keypad, IR sensor, LCD display, LEDs, and a buzzer, into a single functional unit. Through this integration, the system provides a reliable method of access control based on password authentication, while also incorporating essential safety features such as an intruder alarm and system lock mechanism.

Throughout the development process, the project demonstrated the practical application of key microcontroller concepts, such as input and output interfacing, timer usage, and interrupt handling. The use of the Timer0 interrupt allowed the system to manage timing operations efficiently, including idle timeout and system responsiveness. In addition, the implementation of an admin reset function added an extra layer of control and recovery, ensuring that the system could be restored after being locked due to multiple incorrect attempts.

The results obtained from testing confirm that the system operates accurately and reliably under various conditions. It is capable of detecting user presence, processing password input correctly, and responding appropriately to both valid and invalid access attempts. The combination of visual indicators (LEDs and LCD display) and audible alerts (buzzer) further enhances the usability and effectiveness of the system.

Despite its successful implementation, there are several areas where the system can be improved. One possible enhancement is to allow users to change the password dynamically instead of using a fixed predefined password. This would increase flexibility and improve security. Additionally, more advanced authentication methods, such as RFID cards or biometric systems like fingerprint recognition, could be integrated to provide higher levels of protection.

Furthermore, the system could be expanded by incorporating wireless communication technologies such as GSM or WiFi modules. This would enable remote monitoring and real-time alerts, making the system more suitable for modern smart home applications. Another improvement could involve optimizing the software delays to enhance system speed and responsiveness.

Overall, this project has provided valuable experience in designing and implementing an embedded system using the 8051 microcontroller. It highlights how basic components and programming techniques can be combined to create a practical and meaningful real-world application.

5.0 Appendix

```
#include <reg51.h>
#include <intrins.h>

/* FUNCTION PROTOTYPES */
void lcd_send_cmd(unsigned char cmd);
void lcd_send_data(unsigned char dataw);
void lcd_send_str(char *p);
void lcd_write_line(char *str,unsigned char line);
void lcd_init(void);

void intruder_alarm(void);
void buzzer_alarm(void);
char keypad_scan(void);
void show_cursor(void);
void clear_input(void);

/* LCD I2C */
#define LCD_SCL P0^4
#define LCD_SDA P0^5
#define LCD_ADDR 0x4E

sbit scl = LCD_SCL;
sbit sda = LCD_SDA;

/* IR SENSOR */
sbit IR_SENSOR = P3^2;

/* buzzer */
sbit BUZZER = P1^0;

/* LEDs */
sbit LED_GREEN = P1^1;
sbit LED_YELLOW = P1^2;
sbit LED_RED = P1^3;
sbit LED_PURPLE = P1^4;

/* keypad */
sbit R1=P2^0; sbit R2=P2^1; sbit R3=P2^2; sbit R4=P2^3; sbit R5=P2^4; //ROWS
sbit C1=P2^5; sbit C2=P2^6; sbit C3=P2^7; sbit C4=P3^0;           //COLUMNS

unsigned char slave_add;

/* TIMER */
unsigned int timer_tick=0;
unsigned int door_open_count=0;

/* IR system */
bit system_active = 0;
```

```

unsigned int idle_time = 0;
bit idle_displayed = 0;
bit timeout_flag = 0;
bit admin_required = 0;

/* keypad */
char keypad[5][4]={
{'A','B','#','*'},
{'1','2','3','U'},
{'4','5','6','D'},
{'7','8','9','S'},
{'L','0','R','N'}
};

/* password */
char password[4]={'1','2','3','4'};
char input[6];

unsigned char index=0;
unsigned char attempts=0;
unsigned char max_attempt=3;
unsigned char cursor_pos=0;

bit locked=0;
bit admin_mode=0;

/* delay */
void delay_ms(unsigned int n)
{
    unsigned int i,j;
    for(i=0;i<n;i++)
        for(j=0;j<120;j++);
}

/* TIMER */
void timer0_init()
{
    TMOD |= 0x01;
    TH0=0x3C; TL0=0xB0;
    ET0=1;
    EA=1;
}

void timer0_ISR(void) interrupt 1
{
    TH0=0x3C;
    TL0=0xB0;
    timer_tick++;
}

```

```

/* BUZZER */
void buzzer_alarm()
{
    unsigned int i;
    BUZZER=0;
    for(i=0;i<50;i++) delay_ms(100);
    BUZZER=1;
}

/* INTRUDER ALARM SYSTEM */
void intruder_alarm()
{
    unsigned char i;

    for(i=0;i<10;i++)
    {

        /* SCREEN 1 */
        lcd_send_cmd(0x01);
        lcd_write_line("ADMIN FAILED",1);

        LED_RED=1;
        BUZZER=0;

        delay_ms(500);

        LED_RED=0;
        BUZZER=1;

        delay_ms(500);

        /* SCREEN 2 */
        lcd_send_cmd(0x01);
        lcd_write_line("SYSTEM LOCKED",1);
        lcd_write_line("WAIT 30 SEC",2);

        LED_RED=1;
        BUZZER=0;

        delay_ms(500);

        LED_RED=0;
        BUZZER=1;

        delay_ms(500);

    }

    LED_RED=1;

```

```

}

/* I2C */
void i2c_start(){sda=1;scl=1;sda=0;}
void i2c_stop(){scl=0;sda=0;scl=1;sda=1;}
void i2c_ACK(){scl=0;sda=1;scl=1;scl=0;}

void i2c_write(unsigned char dat)
{
    unsigned char i;

    for(i=0;i<8;i++)
    {
        scl=0;
        sda=(dat&(0x80>>i))?1:0;
        scl=1;
    }

    scl=0;
}

/* LCD */
void lcd_slave(unsigned char slave)
{
    slave_add=slave;
}

void lcd_send_cmd(unsigned char cmd)
{
    unsigned char cmd_l,cmd_u;

    cmd_l=(cmd<<4)&0xF0;
    cmd_u=cmd&0xF0;

    i2c_start();
    i2c_write(slave_add);
    i2c_ACK();

    i2c_write(cmd_u|0x0C); i2c_ACK(); delay_ms(1);
    i2c_write(cmd_u|0x08); i2c_ACK(); delay_ms(1);

    i2c_write(cmd_l|0x0C); i2c_ACK(); delay_ms(1);
    i2c_write(cmd_l|0x08); i2c_ACK(); delay_ms(2);

    i2c_stop();
}

void lcd_send_data(unsigned char dataw)
{
    unsigned char data_l,data_u;

```



```

    data_l=(dataw<<4)&0xF0;
    data_u=dataw&0xF0;

    i2c_start();
    i2c_write(slave_add);
    i2c_ACK();

    i2c_write(data_u|0x0D); i2c_ACK(); delay_ms(1);
    i2c_write(data_u|0x09); i2c_ACK(); delay_ms(1);

    i2c_write(data_l|0x0D); i2c_ACK(); delay_ms(1);
    i2c_write(data_l|0x09); i2c_ACK(); delay_ms(2);

    i2c_stop();
}

void lcd_send_str(char *p)
{
    while(*p)
        lcd_send_data(*p++);
}

void lcd_init()
{
    delay_ms(20);

    lcd_send_cmd(0x02);
    delay_ms(5);

    lcd_send_cmd(0x28);
    delay_ms(2);

    lcd_send_cmd(0x0C);
    delay_ms(2);

    lcd_send_cmd(0x06);
    delay_ms(2);

    lcd_send_cmd(0x01);
    delay_ms(5);
}

void lcd_write_line(char *str,unsigned char line)
{
    if(line==1)
        lcd_send_cmd(0x80);
    else
        lcd_send_cmd(0xC0);
}

```

```

        lcd_send_str(str);
    }

/* cursor */
void show_cursor()
{
    if(!locked)
    {
        lcd_send_cmd(0xC0+cursor_pos);
        lcd_send_data('_');
    }
}

/* clear input */
void clear_input()
{
    unsigned char i;
    for(i=0;i<6;i++)
        input[i]=0;
}

/* keypad scan */
char keypad_scan()
{
    char row;

    R1=R2=R3=R4=R5=1;

    for(row=0;row<5;row++)
    {
        R1=R2=R3=R4=R5=1;

        if(row==0)R1=0;
        if(row==1)R2=0;
        if(row==2)R3=0;
        if(row==3)R4=0;
        if(row==4)R5=0;

        if(C1==0){delay_ms(20);while(C1==0);return keypad[row][0];}
        if(C2==0){delay_ms(20);while(C2==0);return keypad[row][1];}
        if(C3==0){delay_ms(20);while(C3==0);return keypad[row][2];}
        if(C4==0){delay_ms(20);while(C4==0);return keypad[row][3];}
    }

    return 0;
}

/* MAIN */

```

```

void main()
{
    char key;
    unsigned char i;

    lcd_slave(LCD_ADDR);
    lcd_init();

    BUZZER = 1;
    LED_GREEN = 0;
    LED_YELLOW = 0;
    LED_RED = 0;
    LED_PURPLE = 0;

    timer0_init();
    TR0=1;

    while(1)
    {
        key = keypad_scan();

        /* IR WAKE */
        if(IR_SENSOR==0 && system_active==0)
        {
            system_active=1;
            idle_time=timer_tick;
            idle_displayed=0;

            timeout_flag = 0;
            admin_required = 0;

            admin_mode = 0;

            LED_PURPLE = 1;

            lcd_send_cmd(0x01);

            if(locked)
            {
                lcd_write_line("SYSTEM LOCKED",1);
                lcd_write_line("CONTACT ADMIN",2);
            }
            else
            {
                lcd_write_line("WELCOME",1);
                lcd_write_line("PLEASE ENTER",2);

                delay_ms(800);

                lcd_send_cmd(0x01);
            }
        }
    }
}

```

```

        lcd_write_line("ENTER PASSWORD",1);

        cursor_pos=0;
        show_cursor();
    }
}

/*NO PEOPLE*/
/* refresh only when key pressed */
if(key!=0 || IR_SENSOR==0)
{
    idle_time = timer_tick;
}

/* timeout (~10s) */
if(timer_tick - idle_time > 200 && !locked)
{
    system_active = 0;
    timeout_flag = 1;

    clear_input();
    index = 0;
    cursor_pos = 0;

    LED_PURPLE = 0;

    admin_mode = 0;
    admin_required = 1;
    attempts = 0;
}

/* IDLE */
if(system_active==0)
{
    if(idle_displayed==0)
    {
        lcd_send_cmd(0x01);

        if(timeout_flag)
        {
            lcd_write_line("TIME OUT",1);
            lcd_write_line("TRY AGAIN",2);
            delay_ms(1500);
            timeout_flag=0;
        }

        lcd_send_cmd(0x01);
        lcd_write_line("SYSTEM READY",1);
        lcd_write_line("WAITING...",2);

        idle_displayed=1;
    }
}

```

```

        }
        continue;
    }

    /* ADMIN / LOCK / NORMAL */
    /* ADMIN MODE */
    if(admin_mode)
    {
        if(key=='S')
        {
            if(index>0)
            {
                index--;
                input[index]=0;
                cursor_pos--;

                lcd_send_cmd(0xC0+cursor_pos);
                lcd_send_data('_');
            }
        }

        else if(key=='A')
        {
            lcd_send_cmd(0xC0);

            for(i=0;i<index;i++)
                lcd_send_data(input[i]);
            delay_ms(1000);
            lcd_send_cmd(0xC0);

            for(i=0;i<index;i++)
                lcd_send_data('*');
            show_cursor();
        }

        else if(key=='N')
        {
            if(index==5 &&
                input[0]=='8'&&
                input[1]=='8'&&
                input[2]=='8'&&
                input[3]=='8'&&
                input[4]=='*')
            {
                locked=0;
                admin_mode=0;
                attempts=0;
            }
        }
    }

```

```

        admin_required = 0;

        LED_RED=0;

        clear_input();

        lcd_send_cmd(0x01);
        lcd_write_line("SYSTEM RESET",1);

        delay_ms(2000);

        lcd_send_cmd(0x01);
        lcd_write_line("ENTER PASSWORD",1);
    }
    else
    {

        intruder_alarm();

        lcd_send_cmd(0x01);
        lcd_write_line("ADMIN PASSWORD",1);
    }

    index=0;
    cursor_pos=0;
    clear_input();
    show_cursor();
}

else if(key!=0)
{
    if(index<5)
    {
        input[index++]=key;

        lcd_send_cmd(0xC0+cursor_pos);
        lcd_send_data("*");

        cursor_pos++;
        show_cursor();
    }
}

else if(admin_required)
{
    if(key=='B')
    {
        admin_mode=1;
        index=0;
        cursor_pos=0;
    }
}

```

```

        lcd_send_cmd(0x01);
        lcd_write_line("ADMIN PASSWORD",1);

        show_cursor();
    }
}
/* NORMAL MODE */
else
{

    /* SHOW DOOR COUNT */

    if(key=='#')
    {
        lcd_send_cmd(0x01);
        lcd_write_line("SYSTEM STATUS",1);

        lcd_send_cmd(0xC0);
        lcd_send_str("DOOR OPEN:");

        lcd_send_data((door_open_count/10)+'0');
        lcd_send_data((door_open_count%10)+'0');

        delay_ms(2000);

        lcd_send_cmd(0x01);
        lcd_write_line("ENTER PASSWORD",1);

        index=0;
        cursor_pos=0;
        clear_input();
        show_cursor();

        continue;
    }
    if(key=='S')
    {
        if(index>0)
        {
            index--;
            input[index]=0;

            cursor_pos--;
            lcd_send_cmd(0xC0+cursor_pos);
            lcd_send_data('_');
        }
    }

    else if(key=='A')

```

```

    {
        lcd_send_cmd(0xC0);

        for(i=0;i<index;i++)
            lcd_send_data(input[i]);

        delay_ms(1000);

        lcd_send_cmd(0xC0);

        for(i=0;i<index;i++)
            lcd_send_data("*");

        show_cursor();
    }

    else if(key=='N')
    {
        if(index<4) continue;

        for(i=0;i<4;i++)
            if(input[i]!=password[i]) break;

        lcd_send_cmd(0x01);

        if(i==4)
        {
            lcd_write_line("ACCESS GRANTED",1);
            LED_GREEN=1;

            attempts=0;
            door_open_count++;

            delay_ms(1500);
            LED_GREEN=0;
        }

        else
        {
            attempts++;

            lcd_write_line("ACCESS DENIED",1);
            LED_YELLOW=1;

            lcd_send_cmd(0xC0);
            lcd_send_str("TRY LEFT:");
            lcd_send_data((max_attempt-attempts)+'0');

            delay_ms(1500);
            LED_YELLOW=0;
        }
    }

```



```

    }

    clear_input();

    if(attempts>=max_attempt)
    {
        locked=1;
        admin_required = 1;

        LED_RED=1;

        lcd_send_cmd(0x01);
        lcd_write_line("SYSTEM LOCKED",1);
        lcd_write_line("CONTACT ADMIN",2);

        buzzer_alarm();

        clear_input();
        index = 0;
        cursor_pos = 0;
    }

    else
    {
        lcd_send_cmd(0x01);
        lcd_write_line("ENTER PASSWORD",1);
    }

    index=0;
    cursor_pos=0;
    show_cursor();
}

else if(key!=0)
{
    if(index<4)
    {
        input[index++]=key;

        lcd_send_cmd(0xC0+cursor_pos);
        lcd_send_data("*");

        cursor_pos++;
        show_cursor();
    }
}

}

}

}

```